PENGYU WANG

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EDUCATION

University of Washington, Michael G. Foster School of Business — Seattle, WA, USA
Ph.D. in Operations Management — Jan. 2023 – expected 2027
Advisors: Prof. Shi Chen, Prof. Yong-Pin Zhou, Prof. Masha Shunko, Prof. Mingwen Yang

University of Washington, Department of Economics — Seattle, WA, USA
Ph.D. in Economics (program; transitioned to OM) — Sept. 2020 – Dec. 2022

University of Wisconsin–Madison, Department of Economics — Madison, WI, USA
Master of Science in Economics — Aug. 2018 – June 2020

Western University — London, Ontario, Canada
Bachelor of Arts, Honors Specialization in Economics — Sept. 2014 – June 2018

RESEARCH INTERESTS

Topics: Sustainable operations, Innovative value chain, Platform operations, AI infrastructure operations, Global supply chain challenges, Semi-conductor industry and supply chain.
Methods: Stochastic optimization, Game theory, Data analysis & Causal inferences, Mean-field games and control, Reinforcement learning.

RESEARCH

(†: Co-first authors).

Working Papers

1. **Wang, P.**, Chen, S., Hou, X. (2026). *Collective ESG Investment: Mitigation, Greenwashing, and Dynamic Strategic Interactions.* Second round revision at *Manufacturing & Service Operations Management* (Aug, 2026).

   - Presented at INFORMS 2025.
   - *Scheduled presentation at INFORMS 2026. **Date**: Tuesday, November 3, 4:15 PM-5:30 PM. **Room**: Moscone South-70 (Lower Mezz).
   - We build a dynamic stochastic framework to study how heterogeneous firms make collective ESG investment decisions, including greenwashing and substantive mitigation, when their actions are strategically interdependent and interact with an ESG-aware financial market.

2. Li, Y.†, **Wang, P.**†, Zhou, Y., Shunko, M. *Can Less Be More? A Hybrid Solution to Print-on-Demand Platforms.* Reject & Resubmit at *Manufacturing & Service Operations Management* (Nov, 2025)

- Accepted by and presented at MSOM conference 2024.
- Studies hybrid make-to-stock / make-to-order (MTS/MTO) production and assortment regulation on print-on-demand platforms. Develops analytical models and contracts to align incentives between platforms and content creators.

3. **Wang, P.**†, Zhang, J.†, Chen, S., Yang, M. (2026). *Platform-Empowered Digital Finance and Traditional Investing: Spillover Effects through Financial Literacy and Risk Tolerance.* Under Review at *Manufacturing & Service Operations Management* (Aug, 2026).

   - *Accepted by and scheduled talk at CIST conference 2026, and DSI conference 2026.
   - This paper studies whether digital finance substitutes for traditional channels or broadens market participation. Using individual-level investment data, we conceptualize digital finance as a gateway that reduces participation frictions, facilitates experiential learning, and promotes migration into adjacent financial markets.

4. **Wang, P.**, Zhou, Y., Shunko, M. *Marketplace Governance and Production Strategy: Decentralized Assortment and Hybrid Fulfillment in Creator Economy.* In preparation for submission.

   - * Scheduled talk at DSI 2026 conferece.
   - We study how creator portfolios are formed in digital marketplaces and the resulting production challenges for platforms. The separation between assortment creation and production decisions creates a need for marketplace governance, which platforms can address through market instruments that reshape creator portfolios and improve performance.

Work in Progress

5. **Wang, P.**, Chen, S. (2026) *A Two-Layer Hybrid Graph Mean-Field Game for Distributed LLM Inference Demand.*

   - Studies the offloading policy and pricing scheme in a data center network that provides AI-computing power.

6. Zong, H., **Wang, P.**, Cao, J., Xu, K. (2026) *Do Drivers Train Their Own Replacement? Dynamic Transition to Robotaxis.*

   - Robotaxi platforms subsidize the human drivers and collect data from them for training the full-self driving (FSD) system. We study whether the FSD system replaces human drivers entirely in a dynamic perspective, and how does the platform's training data collection policy affect the outcomes.

7. **Wang, P.**, Chen, S. (2026) *Global Supply Chain Networks, Dynamics, and Disruption.*

   - Build a theoretical framework to uncover how global supply chain shocks penetrates through the supply network and how firms' endogenous R&D investments accelerate or contain such effects.

8. **Wang, P.**, Zhang, J., Chen, S. (2026). *Who Are the Winners? Business Opportunity Management in High-Tech Industry.*

   - Leverage a unique semi-conductor industry leading company's data to explore how such high-tech company explores business opportunity and foster future business partnership network.

Proceedings

9. **Wang, P.** (2017) *The Causal Relationship between Socioeconomic Status and Smoking Behavior: Evidence from Canada.* Proceedings in Advances in Economics, Business & Management.

CONFERENCES & INVITED TALKS

A Two-Layer Hybrid Graph Mean-Field Game for Distributed LLM Inference Demand:
Decision Science Institute 2026, San Francisco, CA (scheduled)

Marketplace Governance and Production Strategy: Decentralized Assortment and Hybrid Fulfillment in Creator Economy:
Decision Science Institute 2026, San Francisco, CA (scheduled)

Collective ESG Investment: Mitigation, Greenwashing, and Dynamic Strategic Interactions:
INFORMS Annual Meeting 2026, San Francisco, CA (scheduled)
POMS Annual Meeting 2026, Reno, NV
POMS-HK 2026, Shenzhen, China
INFORMS Annual Meeting 2025, Atlanta, GA

Can Less Be More? A Hybrid Solution to Print-on-Demand Platforms:
INFORMS Annual Meeting 2024, Seattle, WA
MOSM 2024 Annual Meeting 2024, Twin-City, MN

Others talks and experiences:
MSOM 2026 Annual Meeting, Boston, MA
SICS Annual Meeting 2025, Berkeley, CA
INFORMS Annual Meeting 2023, Pheonix, AZ

Society Services

Session chairs: INFORMS 2024, MSOM 2024, DSI 2026, INFORMS 2026

TEACHING EXPERIENCE

Invited Instructor: (Evaluations out of 5.0).
IGB 250 Winter 2026, Global Supply Chain: 4.5
IGB 250 Summer 2026, Global Supply Chain: 4.6
Teaching Assistance:
SCM 530A Autumn 2023, Operations Management And Process Analysis: 4.6
SCM 513 Autumn 2023, Managing And Mining Big Data: 4.5.
BUS AN 516 A/B Winter 2024, Operations Research Data Analytics: 4.2.
OPMGT 520A Spring 2024, Operations Management: 4.3

EMBA 533A Winter 2025, Operations Management And Systems Analysis: 4.5
QMETH 505A Spring 2025, Decision Modeling: 4.1;
SCM 520A Spring 2025, Managing Supply Chain Projects: 4.2.

REFERENCES

Prof. Shi Chen
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Prof. Masha Shunko
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